2.21 Hyperbolic Functions: Modelling

Worked Solutions

1 a)

When $x = 0$, $y = 6$:

$$6 = 2p - q\cosh 0 \quad \Rightarrow \quad 6 = 2p - q. \tag{1}$$

When $y = 0$, $x = 3$:

$$0 = 2p - q\cosh 3. \tag{2}$$

Solving (1) and (2) gives $p = 3.33084\ldots = 3.33$ (3 SF) and $q = 0.661692\ldots = 0.662$ (3 SF).

1 b)

$$\begin{aligned} y &= 2(3.33084\ldots) - 0.661692\ldots\cosh x, \\ y\big|_{x=2} &= 6.6616\ldots - 0.661692\ldots\cosh 2 \\ &= 4.1722\ldots. \end{aligned}$$

The vehicle is 4 metres tall and the tunnel is at least 4.17 metres tall (3 SF) if the vehicle drives so that its centre is also at $x = 0$. So yes, the vehicle can pass through the tunnel without touching the sides.

2)

The mean temperature during the first 45 minutes is

$$\begin{aligned} \frac{1}{0.75-0}\int_0^{0.75}(60 - 15\tanh 0.8t)\,dt &= \frac{4}{3}\left[60t - \frac{75}{4}\ln|\cosh 0.8t|\right]_0^{0.75} \\ &= \frac{4}{3}\left(45 - \frac{75}{4}\ln\left|\cosh\frac{3}{5}\right| - (0-0)\right) \\ &= 60 - 25\ln\left|\cosh\frac{3}{5}\right| \\ &= 55.7466\ldots \\ &= 55.75^\circ\text{C} \quad (2\text{ DP}). \end{aligned}$$

3 a)

As $t \to \infty$, $\tanh(0.12t) \to 1$, so $v \to 55 \times 1 = 55$. The model suggests that the terminal velocity is $55\ \text{m}\,\text{s}^{-1}$.

3 b)

90% of the terminal velocity is $0.9 \times 55 = 49.5$.

$$\begin{aligned} &55\tanh(0.12t) = 49.5 \\ \Rightarrow\ &\tanh(0.12t) = 0.9 \\ \Rightarrow\ &0.12t = \operatorname{artanh} 0.9 \\ \Rightarrow\ &t = \frac{25}{3} \times \frac{1}{2}\ln\left(\frac{1+0.9}{1-0.9}\right) \\ &= \frac{25}{6}\ln 19 \\ &= 12.268\ldots \\ &= 12.27 \text{ seconds (2 DP)}. \end{aligned}$$

4 a)

$$\begin{aligned} V &= \pi \int_{-4}^{4} \left(2 + \cosh \frac{x}{2}\right)^2 dx \\ &= \pi \int_{-4}^{4} \left(4 + 4\cosh \frac{x}{2} + \cosh^2 \frac{x}{2}\right) dx. \end{aligned}$$

Since $\cosh^2 A = \frac{1}{2}\cosh 2A + \frac{1}{2}$,

$$\begin{aligned} V &= \pi \int_{-4}^{4} \left(4 + 4\cosh \frac{x}{2} + \frac{1}{2}\cosh x + \frac{1}{2}\right) dx \\ &= \pi \int_{-4}^{4} \left(\frac{9}{2} + 4\cosh \frac{x}{2} + \frac{1}{2}\cosh x\right) dx \\ &= \pi \left[\frac{9}{2}x + 8\sinh \frac{x}{2} + \frac{1}{2}\sinh x\right]_{-4}^{4} \\ &= \pi \left[\left(18 + 8\sinh 2 + \frac{1}{2}\sinh 4\right) - \left(-18 + 8\sinh(-2) + \frac{1}{2}\sinh(-4)\right)\right] \\ &= 381.1370271\ldots. \end{aligned}$$

Therefore $V = 381\,\mathrm{m}^3$ to the nearest cubic metre.

4 b)

The pipe fills at a rate of $0.85\,\mathrm{m}^3\mathrm{s}^{-1}$, so

$$\frac{381}{0.85} = 448.235\ldots.$$

It takes approximately 448 seconds.

5 a)

When $I = fI_0$:

$$\begin{aligned} fI_0 &= I_0 \operatorname{sech}(kx) \\ \Rightarrow f &= \operatorname{sech}(kx) \\ \Rightarrow \frac{1}{f} &= \cosh(kx) \\ \Rightarrow kx &= \operatorname{arcosh}\left(\frac{1}{f}\right) \\ \Rightarrow x &= \frac{1}{k}\operatorname{arcosh}\left(\frac{1}{f}\right). \end{aligned}$$

5 b)

$f = 0.4$ when $x = 12$:

$$\begin{aligned} 12 &= \frac{1}{k}\operatorname{arcosh}\left(\frac{1}{0.4}\right) \\ \Rightarrow k &= \frac{1}{12}\operatorname{arcosh}\frac{5}{2} \\ &= \frac{1}{12}\ln\left(\frac{5}{2} + \sqrt{\left(\frac{5}{2}\right)^2 - 1}\right) \end{aligned}$$

$$= \frac{1}{12} \ln\left(\frac{5}{2} + \frac{\sqrt{21}}{2}\right)$$
$$= 0.1305666\ldots$$
$$\therefore k = 0.131 \quad (3 \text{ SF}).$$

5 c)

$$I_0 = 120, \qquad I = 120\,\text{sech}(kx), \qquad \frac{dI}{dx} = -120k\tanh(kx)\,\text{sech}(kx).$$

5 d)

$$\left.\frac{dI}{dx}\right|_{x=8} = -120 \times 0.131 \tanh(8 \times 0.131)\,\text{sech}(8 \times 0.131)$$
$$= -7.66741\ldots$$
$$= -7.67 \quad (3 \text{ SF}).$$

5 e)

$$\frac{dI}{dx} = -120 \times 0.131 \tanh(0.131x)\,\text{sech}(0.131x)$$
$$= -15.72 \tanh(0.131x)\,\text{sech}(0.131x).$$

For $x \geq 0$, we have $\tanh(0.131x) \geq 0$ and $\text{sech}(0.131x) > 0$, so $\frac{dI}{dx} \leq 0$. As x increases, $\text{sech}(0.131x)$ decreases, and $\tanh(0.131x)$ increases but is bounded from above by 1. So $\tanh(0.131x)\,\text{sech}(0.131x)$ decreases for $x \geq 0$. Therefore $\left|\frac{dI}{dx}\right|$ decreases as x increases for $x \geq 0$. This suggests that the light intensity decreases most rapidly near the surface of the water, and that the rate at which intensity decreases becomes smaller with increasing depth.

5 f)

For small u, $\text{sech}\, u \approx 1 - \frac{u^2}{2}$. So

$$\text{sech}(kx) \approx 1 - \frac{(kx)^2}{2} = 1 - \frac{k^2x^2}{2}$$

for small values of x. Therefore

$$I = 120\,\text{sech}(kx) \approx 120\left(1 - \frac{k^2x^2}{2}\right) = 120 - 60k^2x^2.$$

This approximation suggests that, close to the surface, the intensity of light decreases approximately quadratically with depth. This supports the model's prediction of a rapid initial decrease in intensity as the light enters the water.

6 a)

$$\text{RHS:} \quad \cosh kt + \sinh kt \equiv \frac{e^{kt} + e^{-kt}}{2} + \frac{e^{kt} - e^{-kt}}{2}$$
$$\equiv \frac{1}{2}(e^{kt} + e^{kt}) + \frac{1}{2}(e^{-kt} - e^{-kt})$$
$$\equiv e^{kt} \quad : \text{LHS, as required.}$$

6 b)

$$\begin{aligned}
&\frac{d^2x}{dt^2} - 9x = 0,\\
\text{AE:}\quad &m^2 - 9 = 0 \quad\Rightarrow\quad m = 3 \text{ or } m = -3,\\
&\therefore x = Pe^{3t} + Qe^{-3t}\\
&\quad = P(\cosh 3t + \sinh 3t) + Q(\cosh(-3t) + \sinh(-3t))\\
&\quad = P(\cosh 3t + \sinh 3t) + Q(\cosh 3t - \sinh 3t)\\
&\quad = (P+Q)\cosh 3t + (P-Q)\sinh 3t\\
&\quad = A\cosh 3t + B\sinh 3t, \quad \text{where } A = P+Q,\ B = P-Q.
\end{aligned}$$

When $t = 0$, $x = 2$:

$$2 = A\cosh 0 + B\sinh 0 \quad\Rightarrow\quad A = 2.$$

So $x = 2\cosh 3t + B\sinh 3t$ and

$$v = \frac{dx}{dt} = 6\sinh 3t + 3B\cosh 3t.$$

When $t = 0$, $v = 3$:

$$3 = 6\sinh 0 + 3B\cosh 0 \quad\Rightarrow\quad 3 = 3B \quad\Rightarrow\quad B = 1.$$

Therefore $x = 2\cosh 3t + \sinh 3t$.

6 c)

$$\begin{aligned}
R\cosh(3t+\alpha) &\equiv R(\cosh 3t\cosh\alpha + \sinh 3t\sinh\alpha)\\
&\equiv R\cosh\alpha\cosh 3t + R\sinh\alpha\sinh 3t.
\end{aligned}$$

Thus $R\cosh\alpha = 2$, so $R = \frac{2}{\cosh\alpha}$; and $R\sinh\alpha = 1$, so $R = \frac{1}{\sinh\alpha}$. Hence

$$\begin{aligned}
&\frac{2}{\cosh\alpha} = \frac{1}{\sinh\alpha}\\
\Rightarrow\ &\frac{\sinh\alpha}{\cosh\alpha} = \frac{1}{2}\\
\Rightarrow\ &\tanh\alpha = \frac{1}{2}\\
\Rightarrow\ &\alpha = \operatorname{artanh}\frac{1}{2}\\
&\quad = \frac{1}{2}\ln\left(\frac{1+1/2}{1-1/2}\right) = \frac{1}{2}\ln 3.
\end{aligned}$$

$$\begin{aligned}
R &= \frac{2}{\cosh(\frac{1}{2}\ln 3)} = \frac{2}{\frac{e^{\frac{1}{2}\ln 3} + e^{-\frac{1}{2}\ln 3}}{2}}\\
&= \frac{4}{\sqrt{3} + 1/\sqrt{3}} = \frac{4\sqrt{3}}{3+1} = \sqrt{3}.
\end{aligned}$$

Therefore $x = \sqrt{3}\cosh\left(3t + \frac{1}{2}\ln 3\right)$.

6 d)

$$
\begin{aligned}
2\sqrt{3} &= \sqrt{3}\cosh\left(3t + \frac{1}{2}\ln 3\right) \\
\Rightarrow \cosh\left(3t + \frac{1}{2}\ln 3\right) &= 2 \\
\Rightarrow 3t + \frac{1}{2}\ln 3 &= \operatorname{arcosh} 2 \\
\Rightarrow t &= \frac{1}{3}\operatorname{arcosh} 2 - \frac{1}{6}\ln 3 \\
&= \frac{1}{3}\ln(2 + \sqrt{2^2 - 1}) - \frac{1}{6}\ln 3 \\
&= \frac{1}{3}\ln(2 + \sqrt{3}) - \frac{1}{6}\ln 3 \\
&= \frac{1}{6}\left(2\ln(2 + \sqrt{3}) - \ln 3\right) \\
&= \frac{1}{6}\left(\ln(2 + \sqrt{3})^2 - \ln 3\right) \\
&= \frac{1}{6}\left(\ln(7 + 4\sqrt{3}) - \ln 3\right) \\
&= \frac{1}{6}\ln\left(\frac{7}{3} + \frac{4}{3}\sqrt{3}\right).
\end{aligned}
$$

7 a)

$$
\begin{aligned}
a = \frac{dv}{dt} &= Ak\cosh kt \\
&= Ak\sqrt{\cosh^2 kt} \\
&= Ak\sqrt{1 + \sinh^2 kt} \\
&= k\sqrt{A^2(1 + \sinh^2 kt)} \\
&= k\sqrt{A^2 + A^2\sinh^2 kt} \\
&= k\sqrt{(A\sinh kt)^2 + A^2} \\
&= k\sqrt{v^2 + A^2}.
\end{aligned}
$$

7 b)

When $t = 0$, $a = 2$:

$$
2 = Ak\cosh 0 \quad \Rightarrow \quad Ak = 2 \quad \Rightarrow \quad k = \frac{2}{A}.
$$

When $v = 12$, $a = 5.2$:

$$
\begin{aligned}
5.2 &= k\sqrt{12^2 + A^2} \\
\Rightarrow 5.2 &= \frac{2}{A}\sqrt{144 + A^2} \\
\Rightarrow 2.6A &= \sqrt{144 + A^2} \\
\Rightarrow 6.76A^2 &= 144 + A^2 \\
\Rightarrow 5.76A^2 &= 144
\end{aligned}
$$

$$\Rightarrow A^2 = 25$$
$$\Rightarrow A = 5.$$

7 c)

$$k = \frac{2}{5} = 0.4.$$

7 d)

$$\begin{aligned}\int_0^5 5\sinh(0.4t)\,dt &= \left[\frac{25}{2}\cosh(0.4t)\right]_0^5 \\ &= \frac{25}{2}\cosh 2 - \frac{25}{2}\cosh 0 \\ &= 34.52744\ldots \\ &= 35 \text{ metres to the nearest metre.}\end{aligned}$$

8 a)

$$\begin{aligned}A &= \int_{-6}^{6} \frac{12}{\sqrt{x^2+9}}\,dx \\ &= \int_{-6}^{6} \frac{12}{\sqrt{9(\frac{1}{9}x^2+1)}}\,dx \\ &= \int_{-6}^{6} \frac{12}{3\sqrt{(\frac{1}{3}x)^2+1}}\,dx \\ &= 4\int_{-6}^{6} \frac{1}{\sqrt{(\frac{1}{3}x)^2+1}}\,dx \\ &= 4\left[3\,\text{arsinh}\left(\frac{x}{3}\right)\right]_{-6}^{6} \\ &= 12\ln(2+\sqrt{2^2+1}) - 12\ln(-2+\sqrt{(-2)^2+1}) \\ &= 12\ln(2+\sqrt{5}) - 12\ln(-2+\sqrt{5}) \\ &= 12\ln\left(\frac{2+\sqrt{5}}{-2+\sqrt{5}}\right) \\ &= 12\ln(9+4\sqrt{5}).\end{aligned}$$

Since $9+4\sqrt{5} = (2+\sqrt{5})^2$, then

$$A = 12\ln((2+\sqrt{5})^2) = 24\ln(2+\sqrt{5}).$$

Each unit represents 2 m, so the area scale factor is $2^2 = 4$. Therefore the actual area is

$$4 \times 24\ln(2+\sqrt{5}) = 96\ln(2+\sqrt{5})\,\text{m}^2,$$

as required.

8 b)

$$\begin{aligned} V &= \text{cross-sectional area} \times \text{length} \\ &= 96 \ln(2+\sqrt{5}) \times 18 \\ &= 1728 \ln(2+\sqrt{5})\,\text{m}^3 \\ &= 2495\,\text{m}^3. \end{aligned}$$

8 c)

Each hay bale occupies $1.8\,\text{m}^3$, so the number of bales is

$$\frac{2495}{1.8} = 1386.111\ldots.$$

So 1386 hay bales.